

$$(AB10)_{10} = (11010)_{2} = (0.01101)_{2}$$

$$(0.125)_{10} = (0.001)_{2}$$

Lecture - 1

Number Base Conversion

BA + 20 + Decimal to other num A + 8 = 7

⊗ Decimal to Binary:

$$\begin{aligned} \textcircled{1} (25)_{10} &= (11001)_2 \\ 2 \overline{) 25} &\rightarrow 1 \\ 2 \overline{) 12} &\rightarrow 0 \\ 2 \overline{) 6} &\rightarrow 0 \\ 2 \overline{) 3} &\rightarrow 1 \end{aligned}$$

$$(1.0625)_{10} = (0.25)_{10} = (0.01)_2$$

$$\begin{aligned} 2 \overline{) 3} &\rightarrow 1 \\ 2 \overline{) 1} &\rightarrow 1 \\ 0 & \\ 0.25 \times 2 &= 0.5 \rightarrow 0 \\ 0.5 \times 2 &= 1.0 \rightarrow 1 \end{aligned}$$

$$\textcircled{ii} (58.25)_{10}$$

$$\begin{array}{r} 2 \overline{) 58} \rightarrow 0 \\ 2 \overline{) 29} \rightarrow 1 \\ 2 \overline{) 14} \rightarrow 0 \\ 2 \overline{) 7} \rightarrow 1 \\ 2 \overline{) 3} \rightarrow 1 \\ 2 \overline{) 1} \rightarrow 1 \\ 2 \overline{) 0} \rightarrow 1 \end{array} \uparrow$$

$$\begin{array}{r} 0.25 \times 2 = 0.5 \rightarrow 0 \\ \downarrow \\ 0.5 \times 2 = 1.0 \rightarrow 1 \\ \downarrow \\ 0 \end{array}$$

$$(0.25)_{10} = (0.01)_2$$

$$(58)_{10} = (111010)_2$$

$$\therefore (58.25)_{10} = (111010.01)_2$$

⊗ Decimal to Octal

$$(58.25)_{10} = (72.2)_8$$

$$\begin{array}{r} 8 \overline{) 58} \rightarrow 7 \\ 8 \overline{) 7} \rightarrow 7 \end{array} \uparrow$$

$$\therefore (58.25)_{10} = (72.2)_8$$

$$(0.25)_{10} = (0.2)_8$$

$$\begin{array}{r} 0.25 \times 8 = 2.00 \rightarrow 2 \\ \downarrow \\ 0.00 \end{array}$$

⊗ Decimal to Hexa-Decimal

$$(58.25)_{10} = (3A.4)_{16}$$

$$\begin{array}{r} 16 \overline{) 58} \rightarrow 3 \\ 16 \overline{) 6} \rightarrow 3 \end{array} \uparrow$$

$$(0.25)_{10} = (0.4)_{16}$$

$$\begin{array}{r} 0.25 \times 16 = 4.00 \rightarrow 4 \\ \downarrow \\ 0.00 \end{array}$$

$$\therefore (58.25)_{10} = (3A.4)_{16}$$

Decimal number एक एक पाठान num convert करत, पुन अभागा शून जाग 2 दिता. दशमिक शून 2 द्वारा गुन।

Binary to other number

⊗ Binary to Decimal

$$\textcircled{1} (10110.001)_2 = (?)_{10}$$

$$(10110)_2 \rightarrow \text{binary} = \begin{matrix} 1 & 0 & 1 & 1 & 0 \\ \text{power}(2) = 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{matrix}$$

$$= 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 16 + 4 + 2 = 22$$

$$(0.001)_2 \rightarrow \text{binary} = \begin{matrix} 0 & 0 & 1 \\ (2) = 2^{-1} & 2^{-2} & 2^{-3} \end{matrix}$$

$$= 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 1 \times \frac{1}{8} = 0.125$$

$$\therefore (10110.001)_2 = (22.125)_{10}$$

⊗ Binary to Octal : (3 bit)

$$\textcircled{1} (111110101011.0011)_2 = (?)_8$$

$$\begin{array}{cccccccccccc} 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & . & 0 & 0 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & . & 0 & 0 & 1 & 1 & 0 & 0 \\ \hline 7 & 6 & 5 & 3 & 1 & 4 \end{array}$$

$$(7653.14)_8$$

2	1	0
2	2	2
↓	↓	↓
4	2	4

[दशमिक आरंभ करने के लिए 0 का उपयोग करें। यदि 0 नहीं है तो 0 का उपयोग करें।]

* Binary to Hexadecimal: (4 bit)

$$(10110101011.0011)_2 = (?)_{16}$$

3	2	1	0
2	2	2	2
↓	↓	↓	↓
8	4	2	1

$$\begin{array}{ccccccc} 8 & 4 & 2 & 1 & 8 & 4 & 2 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & . & 0 & 0 & 1 & 1 \\ \hline & 5 & & 10 & & 11 & & 3 & & & & & & \end{array}$$

$$(5A0.3)_{16}$$

0-9	
10 → A	13 → D
11 → B	14 → E
12 → C	15 → F

[দশমিকের আড়া পর 4 bit বা এর count হবে।]

Octal to other number

* Octal to Binary: (3 bit)

$$(152.25)_8 = (?)_2$$

$$\begin{array}{ccccccc} 1 & 5 & 2 & . & 2 & 5 \\ 001 & 101 & 010 & . & 010 & 101 \end{array}$$

$$(001101010.010101)_2$$

আলাদা আলাদা digit কর, প্রতিটা digit এর binary number হবে করতে হবে।

* Octal to Decimal

$$(152.25)_8 = (?)_{10}$$

$$(152)_8 \rightarrow \text{digit} = \begin{array}{ccc} 1 & 5 & 2 \\ (8) & 8^2 & 8^1 & 8^0 \end{array}$$

$$= 1 \times 8^2 + 5 \times 8^1 + 2 \times 8^0 \Rightarrow 64 + 40 + 2 = 106$$

$$(1.25)_8 \rightarrow \text{digit} = \frac{2}{8^1} + \frac{5}{8^2} = \frac{2}{8} + \frac{5}{64} = \frac{16}{64} + \frac{5}{64} = \frac{21}{64}$$

$$= 2 \times 8^{-1} + 5 \times 8^{-2} = 0.25 + 0.078125 = 0.328125$$

$$\therefore (152.25)_8 = (106.328125)_{10} \quad \text{or } (6.0A2)$$

⊗ Octal to Hexadecimal (4 bit)

$$3 \text{ bit } (152.25)_8 = (?)_{16}$$

→ 1st binary convert કરતો રહ્યો, આનામાં આનામાં digit રહે

Digit (001101010.010101)₂

$$(152.25)_8 = (?)_{16}$$

$$(6A.54)_{16}$$

Hexa-decimal to others Number

⊗ Hexa-decimal to Binary: (4 bit)

$$(152A.25)_{16} = (?)_2$$

$$1 \quad 5 \quad 2 \quad A \quad . \quad 2 \quad 5$$

$$(0001010100101010.00100101)_2$$

* Hexa-decimal to Decimal:

$$(152A.25)_{16} = (?)_{10}$$

$$(152A) \rightarrow \text{digit} = 1 \quad 5 \quad 2 \quad A(10)$$

$$(16) = 16^3 \quad 16^2 \quad 16^1 \quad 16^0$$

$$= 1 \times 16^3 + 5 \times 16^2 + 2 \times 16^1 + 10 \times 16^0$$

$$= 4096 + 1280 + 32 + 10$$

$$= 5418$$

$$(25) \rightarrow \text{digit} = 2 \quad 5$$

$$(16) = 16^{-1} \quad 16^{-2}$$

$$= 2 \times 16^{-1} + 5 \times 16^{-2}$$

$$= 0.125 + 0.15625 = 0.28125$$

$$(5418.28125)_{10} \text{ (Ans.)}$$

* Hexa-decimal to Octal:

$$(152A.25)_{16} = (?)_8$$

1st digit - 5A (4 bit) binary number

$$(0001010100101010.00100101)_2$$

(count 3 bit)

000001 010100 101010 001001010
 $\underbrace{\hspace{1.5cm}}_1$ $\underbrace{\hspace{1.5cm}}_2$ $\underbrace{\hspace{1.5cm}}_4$ $\underbrace{\hspace{1.5cm}}_5$ $\underbrace{\hspace{1.5cm}}_2$ $\underbrace{\hspace{1.5cm}}_1$ $\underbrace{\hspace{1.5cm}}_1$ $\underbrace{\hspace{1.5cm}}_2$ (A 521)

$$(12452, 112)_8 \rightarrow 1 \times 8^4 + 2 \times 8^3 + 4 \times 8^2 + 5 \times 8^1 + 2 \times 8^0 = 10144 + 1024 + 256 + 40 + 2 = 11466_{10}$$

Lec-2

Getray Code

⑧ Binary to Gray code:

Q2 Conversion - 4 XOR gate use 24

$\oplus$ XOR = $\left\{ \begin{array}{l} \text{Same Input} \rightarrow 0 \\ \text{Different Input} \rightarrow 1 \end{array} \right\}$

$$0, 0 = 0$$
$$1, 10 = 0$$
$$0,1 = 1$$
 $1, 0 \vdash 1$

① $(1001)_2 = (?)_{10} (1101)_{10}$

$$\begin{array}{ccccccc} & & \uparrow & & & & \\ & & \swarrow & \downarrow & \searrow & & \\ \underline{1} & 0 & 0 & 1 & & & \\ G_4 & G_3 & G_2 & G_1 & & & \end{array}$$
$$G_{xy} \neq 1$$
$$G_3 \neq 1 \oplus 0 = 1 \checkmark$$
$$G_2 \ni 0 \oplus 0 = 0,$$
$$G_{\mathbb{F}_2} \rightarrow 0 \oplus 1 \cong 1$$

⑪ $(0101)_2 = (?)_{Gr} (0111)_{Gray}$

 $G_4 \geq 0$
$$G_2 \rightarrow 0 \oplus 1 = 1$$
$$G_2 \Rightarrow 1 \oplus 0 = 1$$
$$G_{Y \rightarrow 0} \oplus 1 = 1$$
$$B_4 = B_4$$
$$G_3 = B_4 \oplus B_3$$
$$G_2 = B_3 \oplus B_2$$
$$G_1 = B_2 \oplus B_1$$

Gray to Binary :

① $(1010)_G = (?)_2$ $(1100)_2$

$$\begin{array}{cccc} G_4 & G_3 & G_2 & G_1 \\ 1 & 0 & 1 & 0 \end{array}$$

$$B_4 \Rightarrow G_4 = 1$$

$$B_3 \Rightarrow B_4 \oplus G_3 = 1 \oplus 0 = 1$$

$$B_2 \Rightarrow B_3 \oplus G_2 = 1 \oplus 1 = 0$$

$$B_1 \Rightarrow B_2 \oplus G_1 = 0 \oplus 0 = 0$$

$\left\{ \begin{array}{l} \text{same input} \rightarrow 0 \\ \text{differe. n} \rightarrow 1 \end{array} \right\}$

② $(0111)_G = (?)_2$ ~~$(1010)_2$~~

$$\begin{array}{cccc} 0 & 1 & 1 & 1 \end{array}$$

$$B_4 \Rightarrow 10$$

$$B_3 \Rightarrow 1 \oplus 1 = 01$$

$$B_2 \Rightarrow 1 \oplus 1 = 00$$

$$B_1 \Rightarrow 1 \oplus 1 = 01$$

$$(0101)_2$$

$$(0110)_2 = 01(2)$$

Excess-3 Code $\rightarrow$ (4 bit)

Binary to Excess-3

① $(0110)_2 = (?)_{\text{Excess-3}}$

4 bit binary number
নিত ২৮৮

$\rightarrow$ 1st, binary to decimal convert: $0 = 0 + 0$

$$(0110)_2 = (6)_{10}$$

$\rightarrow$ 2nd, decimal num. প্রতিটি digit ক্রমে 3 যোগ।

$$6 + 3 = 9$$

$\rightarrow$ 3rd, প্রতিটি দশকের Binaray হবে ক্রমে।

$$(9)_{10} = (1001)_2$$

$\therefore$ Excess-3 code is $(1001)_2$

⑧ একক অধিক digit থাকলে প্রতিটা digit আলাদা করতে হবে। এরপর প্রতিটা digit এর সাথে 3 যোগ করে binary হবে করতে হবে। এরপর Excess-3 code এর জন্য অবশ্যই digit এর binary প্রকাশ করতে হবে।

⑨ $(31)_{10} = (?)_{\text{Excess}}$

$3+3=6 ; 1+3=4$

$(6)_{10} = (0110)_2 \quad (4)_{10} = (0100)_2$

$(0110 \ 0100)_{\text{Excess}}$

(tip) এখানে 8-সংখ্যক

⑩ $(81.61)_{10} = (?)_{\text{Excess}}$

$8+3=11 ; 1+3=4$

$6+3=9 ; 1+3=4$

$(8)_{10} = (11)_{10} = (1011)_2 \quad (4)_{10} = (0100)_2$

$(6)_{10} = (1001)_2 \quad (4)_{10} = (0100)_2$

$(1011 \ 0100.1001 \ 0100)_{\text{Excess}}$

(tip) এখানে 16-সংখ্যক

$(1001)_{10} = (C)_{16}$

$(1001)_{10}$ is 9-সংখ্যক